

46750 - Optimization in Modern Energy Systems

Assignment 1

Deadline: October 24, 2023 (12:01am)

Instructions: This assignment evaluates the topics covered in **Lectures 1 - 7** as well as programming and writing skills. It should be carried out in groups. Each group should provide a single submission, including the following:

- A concise project report (10 - 14 pages), detailing the mathematical models developed and presenting and analysing the main results. An appendix can be included (outside of page limit).
- A participation table accompanying the report, detailing the participation of each group member (as provided in Lecture 1)
- A working and well-documented code in the programming language of your choice.
- Additional relevant files and data.

All relevant files should be uploaded on DTU Learn. This assignment will count towards 50% of the final grade. The assessment will be based on the grading guide provided.

1. Single-time Step Optimal Power Flow (OPF)

We consider the following power system, based on a modified IEEE 24-node reliability test system, including 12 conventional generating units, 6 wind farms of 300 MW each (at nodes $n_3, n_5, n_7, n_{16}, n_{21}, n_{23}$), 17 demands, and 34 transmission lines, as illustrated in Figure 1. Data on the techno-economic characteristics of the generating units, transmission lines, and load profiles is provided in [1]. Please consider that the capacity of the transmission lines connecting the node pairs (15, 21), (14, 16) and (13, 23) is reduced to 400 MW, 250 MW and 250 MW, respectively. You can assume the production cost of wind farms is zero. Wind data can be found at [2], which can be normalized for your case study. All loads are considered inflexible, however, a load-shedding variable (one for each load) can be introduced, considering a load-shedding cost sufficiently high compared to the production costs of the generators.

In lecture 4 you learned about the DC-OPF problem.

- (a) Please write the optimal flow for a single hour, in a compact way, and then derive its dual formulation and the corresponding KKT conditions.
- (b) Please solve this OPF and compute the values for the locational marginal prices (LMPs), the production cost, and the profit of every supplier.

2. Decomposition of OPF

In Lectures 6 and 7 you learned about decomposition methods for solving large-scale optimization problems.

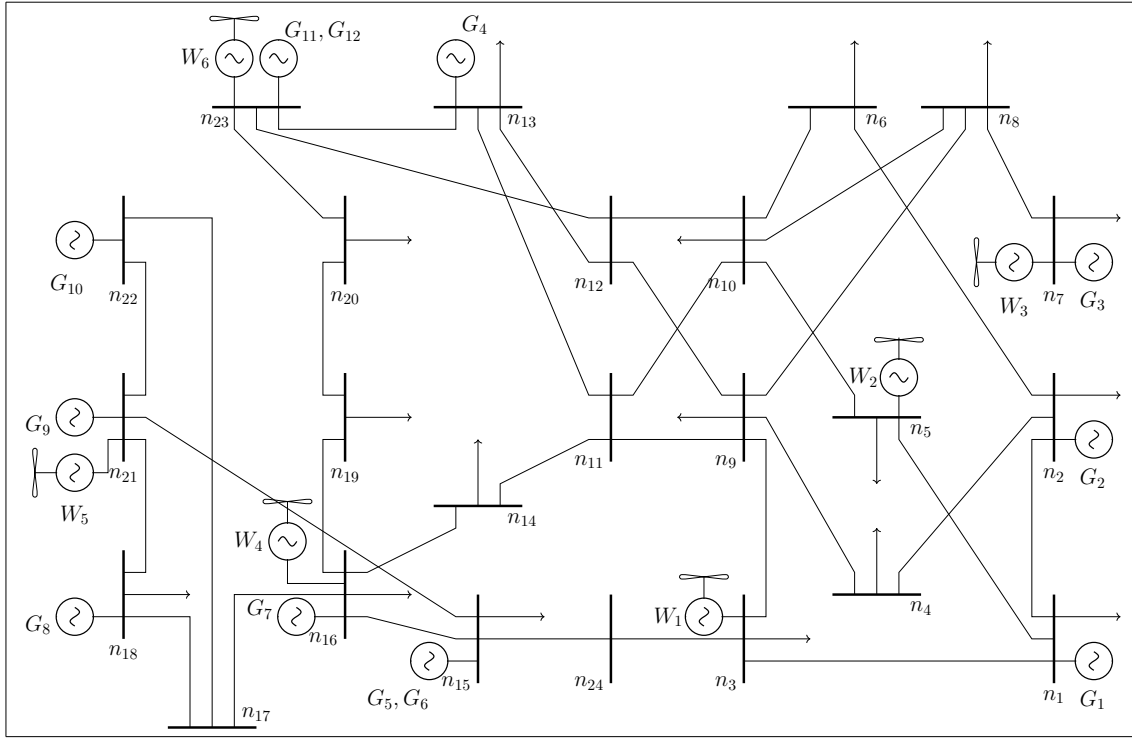


Figure 1: Modified IEEE 24-node electricity system with 6 wind farms

- (a) Please, discuss whether the structure of the single-time step OPF in Question 1 can be decomposed, and if so, in how many subproblems.
- (b) If this problem can be decomposed, please select and implement a suitable decomposition algorithm. Provide an outline of the proposed algorithm, and a detailed formulation of the different steps.
- (c) Present and analyse results about the convergence, and computational tractability of the algorithm.

3. Multi-time Step OPF with Battery Energy Storage Systems (BESSs)

Please, consider now that 3 of the 6 wind farms can be equipped with battery energy storage systems (BESSs), with a maximum storage capacity of 450 MWh each, a charging (discharging) power of 150 MW, and a charging and discharging efficiency of 0.9 and 1.1, respectively. You can consider the operating costs of these BESSs to be 0. You can select arbitrarily the placement, and initial state of charge of these BESSs.

In Lecture 5 you learned how to generalize the formulation of the single-hour OPF with inter-temporal constraints, such as BESSs operation.

- (a) Please extend the formulation of the single-hour OPF in Question 1(a) for 24 hours with these 3 energy storage units. Provide a compact formulation of this multi-time step OPF. The dual formulation and KKT conditions do not need to be repeated in this question.
- (b) Please solve this multi-time step OPF with and without energy storage units, and compare the solutions. How does the introduction of these BESSs impact the LMPs, production cost, congestion, and wind utilization in the system?

4. Optimal Placement of BESSs

As a central planner/system operator, we now consider how to optimally place these 3 BESSs in the power system in order to minimize the system's production cost. Each BESS cannot be split between multiple locations in the power system, however, multiple BESSs can be placed in the same location. As a result the storage capacity installed at each node of the system is either 0 MWh (0 BESS), 450 MWh (1 BESS), 900 MWh (2 BESSs) or 1350 MWh (3 BESSs). The system operation can be modelled over a single (representative) day.

- (a) Please formulate this optimal placement problem for BESSs. specify the decision variables and their type. Is this optimization problem convex?
- (b) Please solve this optimal placement problem and analyze the solutions.

References

- [1] C. Ordoudis, P. Pinson, J. M. Morales, and M. Zugno, "An updated version of the IEEE RTS 24-bus system for electricity market and power system operation studies," *Technical University of Denmark*, vol. 13, 2016.
- [2] W. Bukhsh, "Data for stochastic multiperiod optimal power flow problem," 2025. [Online]. Available: <https://sites.google.com/site/datasmopf/>



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Correction

The numerical results presented below are derived from the assignment of several groups, and provide an example of the type of illustrations and discussions expected.

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1 Nomenclature

We first introduce the sets, input parameters and decision variables of the single-time step DC optimal power flow (OPF) below: **Sets and ranges:**

$\mathcal{G}$ Set of all generators (synchronous and wind)

$\mathcal{D}$ Set of all loads

$\mathcal{N}$ Set of all nodes

$\mathcal{G}_n$ Set of all generators at node $n \in \mathcal{N}$

$\mathcal{D}_n$ Set of all loads at node $n \in \mathcal{N}$

$\mathcal{N}_n$ Set of all nodes connected to node $n \in \mathcal{N}$

$\mathcal{T}$ Set of all time steps ($= 1, \dots, 24$)

$\mathcal{B}$ Set of all batteries ($= 1, 2, 3$)

Input parameters:

$\overline{P}_g^G$ Maximum capacity of generator $g \in \mathcal{G}$ (MW)

$\overline{P}_{dt_0}^D$ Maximum (inflexible) demand of load $d \in \mathcal{D}$ (MW)

$\overline{P}_{nm}^F$ Maximum power flow between nodes $n \in \mathcal{N}$ and $m \in \mathcal{N}_n$, with $\overline{P}_{nm}^F = \overline{P}_{mn}^F$ (MW)

c_g Production cost of generator $g \in \mathcal{G}$ (\$/MWh)

VoLL Value of lost load (\$/MWh)

b_{nm} Susceptance of line connecting nodes $n \in \mathcal{N}$ and $m \in \mathcal{N}_n$, with $b_{nm} = b_{mn}$ (ohm^{-1})

Δ_T Time interval, equal to 1 hour (h)

$\overline{P}_g^G$ Maximum capacity of generator $g \in \mathcal{G}$ (MW)

$\overline{P}_{dt}^D$ Maximum (inflexible) demand of load $d \in \mathcal{D}$ at time $t \in \mathcal{T}$ (MW)

- $\overline{P}_i^{\text{ch}}$ Maximum charging power of battery $i \in \mathcal{B}$ (MW)
- $\overline{P}_i^{\text{dis}}$ Maximum discharging power of battery $i \in \mathcal{B}$ (MW)
- $\overline{\text{SOC}}_i$ Maximum state-of-charge of battery $i \in \mathcal{B}$ (MWh)
- $\text{SOC}_i^{\text{init}}$ Initial state-of-charge of battery $i \in \mathcal{B}$ (MWh)

2 Single-time Step Optimal Power Flow (OPF)

2.1 Primal Formulation

All sets and input parameters the single-time step DC optimal power flow (OPF) at time $t_0 \in \mathcal{T}$ are defined in Section 1. The set of decision variables of the single-time step OPF $\Omega_{t_0}^P = \{\mathbf{p}_{gt_0}\}_{g \in \mathcal{G}} \cup \{\boldsymbol{\theta}_{nt_0}\}_{n \in \mathcal{N}} \cup \{\mathbf{sh}_{dt_0}\}_{d \in \mathcal{D}}$ includes the production $\mathbf{p}_{gt_0}$ of each generator $g \in \mathcal{G}$ (including wind producers), the voltage angle $\boldsymbol{\theta}_{nt_0}$ at each node $n \in \mathcal{N}$, and load shedding $\mathbf{sh}_{dt_0}$ for each load $d \in \mathcal{D}$. This single-time step OPF at time $t_0 \in \mathcal{T}$ is formulated as

$$\min_{\Omega_{t_0}^P} \sum_{g \in \mathcal{G}} c_g \mathbf{p}_{gt_0} + \sum_{d \in \mathcal{D}} \text{VoLL} \mathbf{sh}_{dt_0} \quad (1a)$$

$$\begin{aligned} \text{s.t.} \quad \sum_{g \in \mathcal{G}_n} \mathbf{p}_{gt_0} &= \sum_{d \in \mathcal{D}_n} (\bar{P}_{dt_0} - \mathbf{sh}_{dt_0}) \\ &+ \sum_{m \in \mathcal{N}_n} b_{nm} (\boldsymbol{\theta}_{nt_0} - \boldsymbol{\theta}_{mt_0}), \quad \forall n \in \mathcal{N} \quad : \boldsymbol{\lambda}_{nt_0} \end{aligned} \quad (1b)$$

$$0 \leq \mathbf{p}_{gt_0} \leq \bar{P}_g^G, \quad \forall g \in \mathcal{G} \quad : \underline{\boldsymbol{\mu}}_{gt_0}^G, \bar{\boldsymbol{\mu}}_{gt_0}^G \quad (1c)$$

$$0 \leq \mathbf{sh}_{dt_0} \leq \bar{P}_{dt_0}^D, \quad \forall d \in \mathcal{D} \quad : \underline{\boldsymbol{\mu}}_{dt_0}^D, \bar{\boldsymbol{\mu}}_{dt_0}^D \quad (1d)$$

$$b_{nm} (\boldsymbol{\theta}_{nt_0} - \boldsymbol{\theta}_{mt_0}) \leq \bar{P}_{nm}^F, \quad \forall n \in \mathcal{N}, m \in \mathcal{N}_n \quad : \bar{\boldsymbol{\mu}}_{nm}^F \quad (1e)$$

$$\boldsymbol{\theta}_{1t_0} = 0 \quad : \boldsymbol{\sigma}_{1t_0}, \quad (1f)$$

which aims at minimizing the system operating cost (1a), consisting of electricity production cost and load shedding cost, subject to constraints on the power balance at each node (1b), upper- and lower-bounds on the power production of each generator (1c), load shedding of each consumer (1d), and power flow between all connected nodes (1e), as well as the reference voltage angle at the slack bus (1f). Note that, in this formulation, the constraint (1e) are defined twice for each pair (n, m) of connected nodes. As a result, for a given $n \in \mathcal{N}$ and $m \in \mathcal{N}_n$, the flow $b_{nm} (\boldsymbol{\theta}_{nt_0} - \boldsymbol{\theta}_{mt_0})$ between nodes n and m is upper-bounded by $\bar{P}_{nm}^F$, and lower bounded by $-\bar{P}_{mn}^F$, with $b_{nm} (\boldsymbol{\theta}_{nt_0} - \boldsymbol{\theta}_{mt_0}) > 0$ if the power flows from node n to node m . The dual variables associated with each constraint are defined in front of that constraint. These dual variables represent the sensitivity of the objective function of the OPF problem at optimality, with respect to the right-hand-side of

the constraints they are associated with. In particular, λ_{nt_0} represents the nodal price at node n . This minimization problem is convex, because its objective function, equality and inequality constraints are all **linear**, and therefore convex.

2.2 Dual Formulation

The dual formulation of the single-time sep OPF at time t_0 is as follows:

$$\max_{\Omega^D} \sum_{n \in \mathcal{N}} \sum_{d \in \mathcal{D}_n} \bar{P}_{dt_0}^D \lambda_{nt_0} - \sum_{g \in \mathcal{G}} \bar{P}_g^G \bar{\mu}_{gt_0}^G - \sum_{d \in \mathcal{D}} \bar{P}_{dt_0}^D \bar{\mu}_{dt_0}^D - \sum_{n \in \mathcal{N}} \sum_{m \in \mathcal{N}_n} \bar{P}_{nm}^F \bar{\mu}_{nmt_0}^F \quad (2a)$$

$$\text{s.t. } \lambda_{nt_0} - \bar{\mu}_{gt_0}^G \leq c_g, \forall n \in \mathcal{N}, g \in \mathcal{G}_n \quad : \mathbf{p}_{gt_0} \quad (2b)$$

$$\lambda_{nt_0} - \bar{\mu}_{dt_0}^D \leq VoLL, \forall n \in \mathcal{N}, d \in \mathcal{D}_n \quad : \mathbf{sh}_{dt_0} \quad (2c)$$

$$\sum_{m \in \mathcal{N}_n} b_{nm} (\lambda_{nt_0} - \lambda_{mt_0} + \bar{\mu}_{nmt_0}^F - \bar{\mu}_{mnt_0}^F) + \delta_1(n) \sigma_{1t_0} = 0, \forall n \in \mathcal{N} \quad : \boldsymbol{\theta}_{nt_0} \quad (2d)$$

$$\bar{\mu}_{gt_0}^G \geq 0, \forall g \in \mathcal{G} \quad (2e)$$

$$\bar{\mu}_{dt_0}^D \geq 0, \forall d \in \mathcal{D} \quad (2f)$$

$$\bar{\mu}_{nmt_0}^F \geq 0, \forall n \in \mathcal{N}, m \in \mathcal{N}_n \quad (2g)$$

with the set of dual decision variables $\Omega_{t_0}^D = \{\lambda_{nt_0}\}_{n \in \mathcal{N}} \cup \{\bar{\mu}_{gt_0}^G\}_{g \in \mathcal{G}} \cup \{\bar{\mu}_{dt_0}^D\}_{d \in \mathcal{D}} \cup \{\bar{\mu}_{nmt_0}^F\}_{n \in \mathcal{N}, m \in \mathcal{N}_n} \cup \{\sigma_{1t_0}\}$ including all dual variables associated with constraints (1b)-(1f). Constraints (2b)-(2d) are the dual constraints associated with the primal variables indicated in front of them, and (2e)-(2g) are the dual feasibility constraints. Note that the parameter $\delta_1(n)$ in (2d) is equal to 1 if $n = 1$ and 0 otherwise.

2.3 KKT Conditions

We formulate the Lagrange function of this single time step OPF at time t_0 as follows:

$$\begin{aligned}
\mathcal{L} = & (1a) + \sum_{n \in \mathcal{N}} \lambda_{nt_0} \left[\sum_{d \in \mathcal{D}_n} (\bar{P}_{dt_0} - \mathbf{s}h_{dt_0}) + \sum_{m \in \mathcal{N}_n} b_{nm} (\theta_{nt_0} - \theta_{mt_0}) - \sum_{g \in \mathcal{G}_n} p_{gt_0} \right] \\
& + \sum_{g \in \mathcal{G}} \left\{ -\underline{\mu}_{gt_0}^G p_{gt_0} + \bar{\mu}_{gt_0}^G [p_{gt_0} - \bar{P}_g^G] \right\} \\
& + \sum_{d \in \mathcal{D}} \left\{ -\underline{\mu}_{dt_0}^D \mathbf{s}h_{dt_0} + \bar{\mu}_{dt_0}^D [\mathbf{s}h_{dt_0} - \bar{P}_{dt_0}^D] \right\} \\
& + \sum_{n \in \mathcal{N}} \sum_{m \in \mathcal{N}_n} \left\{ \bar{\mu}_{nmt_0}^F [b_{nm} (\theta_{nt_0} - \theta_{mt_0}) - \bar{P}_{nm}^F] \right\} + \sigma_{1t_0} \theta_{1t_0}
\end{aligned} \tag{3}$$

As a result, the KKT conditions of the single-time step OPF at time t_0 are formulated as:

$$\frac{\partial \mathcal{L}}{\partial p_{gt_0}} = c_g - \lambda_{nt_0} - \underline{\mu}_{gt_0}^G + \bar{\mu}_{gt_0}^G = 0, \forall n \in \mathcal{N}, g \in \mathcal{G}_n \tag{4a}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{s}h_{dt_0}} = \text{VoLL} - \lambda_{nt_0} - \underline{\mu}_{dt_0}^D + \bar{\mu}_{dt_0}^D = 0, \forall n \in \mathcal{N}, d \in \mathcal{D}_n \tag{4b}$$

$$\frac{\partial \mathcal{L}}{\partial \theta_{nt_0}} = \sum_{m \in \mathcal{N}_n} b_{nm} (\lambda_{nt_0} - \lambda_{mt_0} + \bar{\mu}_{nmt_0}^F - \bar{\mu}_{mnt_0}^F) + \delta_1(n) \sigma_{1t_0} = 0, \forall n \in \mathcal{N} \tag{4c}$$

$$\text{Constraints (1b), (1f)} \tag{4d}$$

$$0 \leq \bar{\mu}_{gt_0}^G \perp (p_{gt_0} - \bar{P}_g^G) \leq 0, \forall g \in \mathcal{G} \tag{4e}$$

$$0 \leq \underline{\mu}_{gt_0}^G \perp (-p_{gt_0}) \leq 0, \forall g \in \mathcal{G} \tag{4f}$$

$$0 \leq \bar{\mu}_{dt_0}^D \perp (\mathbf{s}h_{dt_0} - \bar{P}_{dt_0}^D) \leq 0, \forall d \in \mathcal{D} \tag{4g}$$

$$0 \leq \underline{\mu}_{dt_0}^D \perp (-\mathbf{s}h_{dt_0}) \leq 0, \forall d \in \mathcal{D} \tag{4h}$$

$$0 \leq \bar{\mu}_{nmt_0}^F \perp (b_{nm} (\theta_{nt_0} - \theta_{mt_0}) - \bar{P}_{nm}^F) \leq 0, \forall n \in \mathcal{N}, m \in \mathcal{N}_n, \tag{4i}$$

where (4a)-(4c) represent the first-order conditions, i.e. the derivatives of the Lagrangian with respect to each primal decision variables p_{gt_0} , θ_{nt_0} and $\mathbf{s}h_{dt_0}$, (4d) and the right side of (4e)-(4i) represent the primal feasibility conditions, i.e. all the constraints of the primal OPF problem (1b)-(1f), the left side of (4e)-(4i) represent the dual feasibility conditions, i.e. non-negativity of all dual variables associated with primal inequality constraints, and the $\perp$ sign in (4e)-(4i) represents the complementarity conditions for all primal inequality constraints. We observe a parallel

between the dual and the KKT conditions of the OPF problem. In particular, (4a)-(4c) are equivalent to (2b)-(2d).

2.4 Numerical Results

We solve this single-time step OPF for the Updated Version of the IEEE RTS 24-Bus System introduced in [1], including 12 conventional generating units, 6 wind farms of 300 MW each (at nodes $n_3, n_5, n_7, n_{16}, n_{21}, n_{23}$), 17 demands, and 34 transmission lines, as illustrated in Figure ?? . Data on the techno-economic characteristics of the generating units, transmission lines, and load profiles is provided in [1]. We consider that the capacity of the transmission lines connecting the node pairs (15, 21), (14, 16) and (13, 23) is reduced to 400 MW, 250 MW and 250 MW, respectively. We assume the production cost of wind farms is zero. We choose to solve this OPF at hour t . Additionally, we use the scenarios of normalized wind production provided in [2], and choose the k^{th} scenario.

The resulting dispatch from generators and wind farms as well as node LMPs are illustrated in fig. 1 while the key figures are summarized in tabs. 1 and 2. The optimal objective value (i.e., the lowest total production cost) using the first hour of wind and demand data is 4754.43 \$. From the figure, it is evident that only generators G8, G9, G10 and G11 are in use while the rest of the generators are idle. Only G10 is fully dispatched and as a result only G10 has a dual variable different from zero as the power limit constraint associated with this variable is binding. If generator G10 could increase its capacity further, the objective value would decrease. It is also only generator G10 which is generating profits. The profits are calculated as the difference between the production cost and the LMP multiplied by the production quantity. Thus, when generators G8, G9, and G11 have zero profits despite being active, it means that they are marginal generators. The costs of generators are found in the description of the techno-economics characteristics for the system [1]. For this system, G10 has a production cost of 0 \$/MWh, which explains why it is fully dispatched.

The dual variables for the maximum flow constraints between the nodes reveal that there is congestion in the system between nodes 14 and 16 and between nodes 15 and 21. This congestion occurs because most of the production occurs in the left side of the network, likely because these generators have the lowest

production costs.

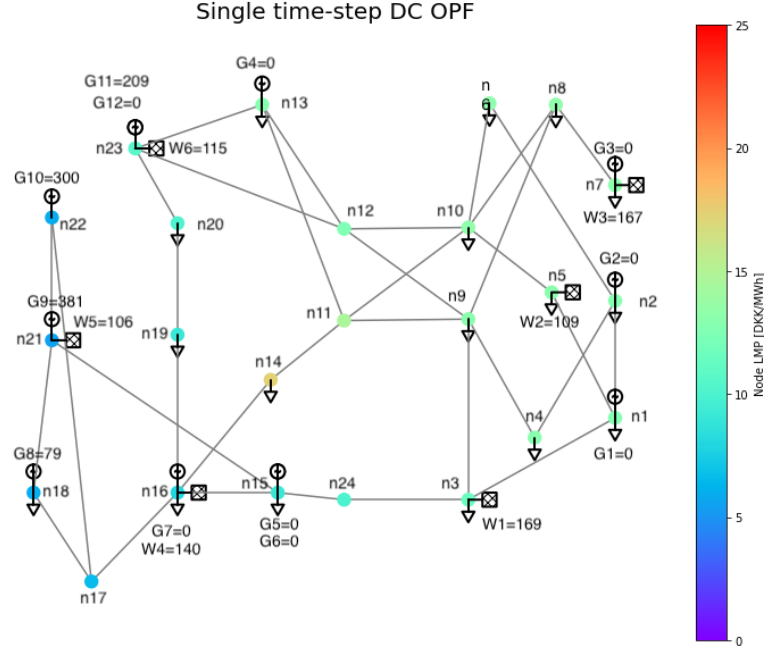


Figure 1: Network LMPs, generator production, and wind production for single time-step DC OPF. Generators are attached to their nodes as black circles, wind farms as squares, and loads as down-pointing arrows.

The locational marginal prices (LMPs) are the dual variables of the balancing constraints for each node, i.e., the increase in objective function if the demand at the node is increased by 1 MWh. Because the cheap generators are located in the left (and there is congestion on two lines connecting the left with the right), the LMPs are significantly lower in the left of the network with values in the range 5-10 \$/MWh while nodes in the right have values of 10-15 \$/MWh. The highest LMP is at node N14 which is at the end of a congested line.

In tab. 2 each wind farm's production and profit can be seen. It is evident that all wind farms are in use and that the wind farms are utilized to their maximum capacity which is expected as the cost of production is assumed to be zero. Consequently, all wind farms generate profits.

Table 1: Variables and parameters related to the Generators

Single-time Step Optimal Power Flow (Generators)												
	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
Production [MWh]	0	0	0	0	0	0	0	79.11	380.63	300	208.76	0
Cost [\$]	0	0	0	0	0	0	0	476.3	2082.1	0	2196.1	0
Profit [\$]	0	0	0	0	0	0	0	0	0	1776.1	0	0

Table 2: Variables and parameters related to the Wind farms

Single-time Step Optimal Power Flow (Wind farms)						
	W1	W2	W3	W4	W5	W6
Production [MWh]	169.01	108.78	167.48	140.46	106.26	115.34
Profit [\$]	1941.93	1420.14	2209.34	1107.19	581.27	1213.36

3 Decomposition of OPF

We consider the single-time step OPF problem at time $t_0 \in \mathcal{T}$ defined in Section 2, and the notations previously introduced in Section (1).

3.1 Decomposable Structure and Decomposition Method

ALR and ADMM: The single time-step OPF in (1a)-(1f) is a linear problem, with a separable objective function, and complicating constraints (1b) (nodal balance equations) and (1e) (power flow bounds) which link the variables at multiple nodes of the system. Therefore, it is possible to decompose it using Augmented Lagrangian Relaxation (ALR) methods, and, in particular, the Alternating Direction Multipliers Method (ADMM). If these constraints are relaxed, the original problem can be decomposed into as many subproblems as there are generators and loads in the system, i.e. $|\mathcal{G}| + |\mathcal{D}|$.

Consensus ADMM: Since each node of the system represents a subsystem that is linked to other subsystems only through the power flow between them (proportional to the difference of voltage angles $\theta_{it_0} - \theta_{kt_0}$ between nodes i and k), the consensus ADMM can be used. Here, local copies θ_{ikt_0} of the global voltage angles θ_{it_0} at each node i are created, and represent the voltage angle at node $i \in \mathcal{N}$

seen from the neighboring node $k \in \mathcal{N}_i$. The operation of each subsystem (node) k in $\mathcal{N}$ can then be modelled using only the local variables θ_{ikt_0} for $i \in \mathcal{N}_k$, p_{gt_0} for $g \in \mathcal{G}_k$ and $\mathbf{sh}_{dt_0}$ for $d \in \mathcal{D}_k$ associated with this node. Consensus between the local copies of the global variables can then be enforced by

$$\theta_{it_0} = \theta_{ikt_0}, \forall k \in \mathcal{N}_i, i \in \mathcal{N}. \quad (5)$$

As a result, replacing the occurrences of the global variables with the local copies yields the following reformulation of the single time-step DC OPF:

$$\min_{\tilde{\Omega}_{t_0}^P} \sum_{g \in \mathcal{G}} c_g p_{gt_0} + \sum_{d \in \mathcal{D}} \text{VoLL} \mathbf{sh}_{dt_0} \quad (6a)$$

$$\text{s.t.} \quad \sum_{g \in \mathcal{G}_i} p_{gt_0} = \sum_{d \in \mathcal{D}_i} (\bar{P}_{dt_0} - \mathbf{sh}_{dt_0}) + \sum_{k \in \mathcal{N}_i} b_{ikt_0} (\theta_{iit_0} - \theta_{kit_0}), \forall i \in \mathcal{N} \quad (6b)$$

$$\text{Equations (1c) -- (1d)} \quad (6c)$$

$$b_{ikt_0} (\theta_{iit_0} - \theta_{kit_0}) \leq \bar{P}_{ik}, \forall i \in \mathcal{N}, k \in \mathcal{N}_i \quad (6d)$$

$$\theta_{1k} = 0, \forall k \in \mathcal{N}_1 \quad (6e)$$

$$\theta_{ikt_0} = \theta_{it_0}, \forall i \in \mathcal{N}, k \in \mathcal{N}_i \cup \{i\} : \eta_{ikt_0}, \quad (6f)$$

where set of decision variables $\tilde{\Omega}_{t_0}^P = \{p_{gt_0}\}_{g \in \mathcal{G}} \cup \{\mathbf{sh}_{dt_0}\}_{d \in \mathcal{D}} \cup \{\theta_{ikt_0}\}_{i \in \mathcal{N}, k \in \mathcal{N}_i} \cup \{\theta_{it_0}\}_{i \in \mathcal{N}}$ includes the generators' production, loads' shedding, nodal local and global voltage angles. This OPF problem is similar to the formulation in (1a)-(1f), where the global voltage angle variables have been replaced by their local copies in all constraints (6b), (6d), and (6e), and constraint (6f) sets the local copies equal to their respective global variable. Note that η_{ikt_0} for nodes $i \in \mathcal{N}, k \in \mathcal{N}_i$ are the dual variables associated with the consensus constraints (6f). In this formulation, the only complicating constraints linking decision variables associated with more than one node is the newly added consensus constraints (6f). We can treat them as decision variables. In this case, by relaxing the consensus constraints (6f), the OPF problem can be decomposed into 2 types of subproblems. The first type of subproblems decides the dispatch of generators and local voltage angles at each node ($|\mathcal{N}|$ subproblems), and the second type of subproblems updates the global voltage angles at each node ($|\mathcal{N}|$ subproblems).

3.2 Consensus ADMM Algorithm

The relaxation of Problem (6a)-(6f) is achieved by relaxing the consensus constraints (6f) and including them in the objective function via the Lagrangian term $\sum_{i \in \mathcal{N}, k \in \mathcal{N}_i} \hat{\eta}_{ikt_0} (\boldsymbol{\theta}_{ikt_0} - \boldsymbol{\theta}_{it_0})$, where $\hat{\eta}_{ikt_0}$ are fixed values of the Lagrange multipliers associated with the complicating constraints (6f), and the quadratic penalty terms $\sum_{i \in \mathcal{N}, k \in \mathcal{N}_i} \frac{\gamma}{2} (\boldsymbol{\theta}_{ikt_0} - \boldsymbol{\theta}_{it_0})^2$, which are equal to zero at optimality, where $\gamma > 0$ is a fixed parameter. The resulting relaxed problem can be decomposed per node and solved iteratively using the consensus ADMM algorithm if in each subproblem at node $i \in \mathcal{N}$ variables that do not belong to this subproblem are fixed. The steps of this consensus ADMM algorithm are detailed below, and summarized in Algorithm 1.

Initialization:

At the initial iteration $\nu = 0$, initial values for all primal variables and Lagrange multipliers of the OPF problem in (6a)-(6f) are chosen arbitrarily.

Subproblem 1:

At each following iteration $\nu \geq 1$ and for each node $i \in \mathcal{N}$, the values of the production $p_{gt_0}^{(\nu)}$ of generators $g \in \mathcal{G}_i$, the load shedding $sh_{dt_0}^{(\nu)}$ of loads $d \in \mathcal{D}_i$, and the local copies of the voltage angles $\theta_{kit_0}^{(\nu)}$ at nodes $k \in \mathcal{N}_i \cup \{i\}$ as seen by node i are determined as the solutions of the 1st type of subproblems (SUB $_i^{(\nu)}$),

formulated as:

$$\begin{aligned} \min_{\Omega_{it_0}^{\text{SUB}}} \quad & \sum_{g \in \mathcal{G}_i} c_g \mathbf{p}_{gt_0} + \sum_{d \in \mathcal{D}_i} \text{VoLL} \mathbf{sh}_{dt_0} \\ & + \sum_{k \in \mathcal{N}_i \cup \{i\}} \left[\eta_{kit_0}^{(\nu-1)} (\boldsymbol{\theta}_{kit_0} - \theta_{kt_0}^{(\nu-1)}) + \frac{\gamma}{2} (\boldsymbol{\theta}_{kit_0} - \theta_{kt_0}^{(\nu-1)})^2 \right] \end{aligned} \quad (7a)$$

$$\text{s.t.} \quad \sum_{g \in \mathcal{G}_i} \mathbf{p}_{gt_0} = \sum_{d \in \mathcal{D}_i} (\bar{P}_{dt_0} - \mathbf{sh}_{dt_0}) + \sum_{k \in \mathcal{N}_i} b_{ikt_0} (\boldsymbol{\theta}_{iit_0} - \boldsymbol{\theta}_{kit_0}) \quad (7b)$$

$$0 \leq \mathbf{p}_{gt_0} \leq \bar{P}_g^G, \quad \forall g \in \mathcal{G}_i \quad (7c)$$

$$0 \leq \mathbf{sh}_{dt_0} \leq \bar{P}_{dt_0}^D, \quad \forall d \in \mathcal{D}_i \quad (7d)$$

$$b_{ikt_0} (\boldsymbol{\theta}_{iit_0} - \boldsymbol{\theta}_{kit_0}) \leq \bar{P}_{ik}, \quad \forall k \in \mathcal{N}_i \quad (7e)$$

$$\boldsymbol{\theta}_{1i} = 0, \quad (7f)$$

$$(7g)$$

where $\Omega_{it_0}^{\text{SUB}} = \{\mathbf{p}_{gt_0}\}_{g \in \mathcal{G}_i} \cup \{\mathbf{sh}_{dt_0}\}_{d \in \mathcal{D}_i} \cup \{\boldsymbol{\theta}_{kit_0}\}_{k \in \mathcal{N}_i \cup \{i\}}$ includes the production of generators and load shedding at this node, and the local copies of the voltage angles as seen by this node. The constraints of this subproblem (7b)-(7f) represent the OPF constraints at node i only. Note that $\eta_{kit_0}^{(\nu-1)}$ and $\theta_{kt_0}^{(\nu-1)}$ are fixed to their values at previous iteration and are not decision variables.

Subproblem 2:

Subsequently, at each iteration $\nu \geq 1$ and for each node $i \in \mathcal{N}$, the values of the global voltage angle variables $\theta_{it_0}^{(\nu)}$ are determined as the solutions of the 2nd type of subproblems ($\tilde{\text{SUB}}_i^{(\nu)}$), formulated as:

$$\min_{\Omega_{it_0}^{\tilde{\text{SUB}}}} \quad \sum_{k \in \mathcal{N}_i \cup \{i\}} \left[\eta_{ikt_0}^{(\nu-1)} (\theta_{ikt_0}^{(\nu-1)} - \boldsymbol{\theta}_{it_0}) + \frac{\gamma}{2} (\theta_{ikt_0}^{(\nu-1)} - \boldsymbol{\theta}_{it_0})^2 \right], \quad (8)$$

where $\Omega_{it_0}^{\tilde{\text{SUB}}} = \{\boldsymbol{\theta}_{it_0}\}$ includes the global voltage angle at this node. Note that $\eta_{ikt_0}^{(\nu-1)}$ and $\theta_{ikt_0}^{(\nu-1)}$ are fixed to their values at previous iteration and are not decision variables. Since this second type of subproblems is unconstrained, its analytical solutions can be derived analytically, and at optimality it will converge to $\boldsymbol{\theta}_{it_0}^{(\nu)} =$

$$\frac{(\sum_{k \in \mathcal{N}_i} \theta_{ikt_0}^{(\nu-1)} + \theta_{iit_0}^{(\nu-1)})}{|\mathcal{N}_i| + 1}.$$

Master Problem:

At each iteration $\nu \geq 1$, after the updates of both subproblems, the values of the Lagrange multipliers η_{kit_0} can now be updated in the master problem, such that:

$$\eta_{kit_0}^{(\nu)} = \eta_{kit_0}^{(\nu-1)} + \gamma \cdot (\theta_{kit_0}^{(\nu)} - \theta_{kit_0}^{(\nu-1)}), \forall i \in \mathcal{N}, k \in \mathcal{N}_i \cup \{i\} \quad (9)$$

Convergence check:

Finally, a convergence check is used to end the algorithm when the changes in $\eta_{kit_0}^{(\nu)}$ from one iteration to the next are deemed sufficiently small, i.e.:

$$\Delta_{\eta_{t_0}}^{(\nu)} = \frac{\|\eta_{t_0}^{(\nu)} - \eta_{t_0}^{(\nu-1)}\|}{\|\eta_{t_0}^{(\nu-1)}\|} \leq \epsilon \quad (10)$$

where we define the vector $\eta_{t_0}^{(\nu)} = [\eta_{ikt_0}^{(\nu)}]_{i \in \mathcal{N}, k \in \mathcal{N}_i \cup \{i\}}$. As evident from eq. (9), a sufficiently small change in η_{kit_0} means the relaxation can be assumed to be exact. If this condition is fulfilled the algorithm terminates. If it is not fulfilled $\nu \leftarrow \nu + 1$ and the subproblems are solved again. This Consensus ADMM algorithm is summarized in algorithm 1 and is implemented in the python function `runConsensusADMM`.

Algorithm 1 Consensus ADMM

Set $\{p_{gt_0}^{(0)}\}_{g \in \mathcal{G}_i}, \{sh_{dt_0}^{(0)}\}_{d \in \mathcal{D}_i}, \{\theta_{kit_0}^{(0)}, \eta_{kit_0}^{(0)}\}_{i \in \mathcal{N}, k \in \mathcal{N}_i}$ and $\Delta_{\eta_{t_0}}^{(0)} > 1$ $\triangleright$ Initialize values of primal variables, Lagrange multipliers, and dual residual.

$\nu \leftarrow 1$

while $e > \epsilon$ & $v < \text{max_iter}$ **do**

for $i \in \mathcal{N}$ **do**

$\{p_{gt_0}^{(\nu)}\}_{g \in \mathcal{G}_i} \cup \{sh_{dt_0}^{(\nu)}\}_{d \in \mathcal{D}_i} \cup \{\theta_{kit_0}^{(\nu)}\}_{k \in \mathcal{N}_i} \leftarrow \text{argmin. (7a) - (7f)}$ $\triangleright$ Solve subproblem $\text{SUB}_i^{(\nu)}$

end for

for $i \in \mathcal{N}$ **do**

$\theta_{it_0}^{(\nu)} \leftarrow \text{argmin. (8)}$ $\triangleright$ Solve subproblem $\tilde{\text{SUB}}_i^{(\nu)}$

end for

for $i \in \mathcal{N}$ **do**

for $k \in \mathcal{N}_i$ **do**

$\eta_{kit_0}^{(\nu)} \leftarrow \eta_{kit_0}^{(\nu-1)} + \gamma \cdot (\theta_{kit_0}^{(\nu)} - \theta_{kt_0}^{(\nu)})$ $\triangleright$ Update Lagrange multipliers.

end for

end for

$\Delta_{\eta_{t_0}}^{(\nu)} \leftarrow \frac{\|\eta_{t_0}^{(\nu)} - \eta_{t_0}^{(\nu-1)}\|}{\|\eta_{t_0}^{(\nu-1)}\|}$ $\triangleright$ Compute dual residual.

$\nu \leftarrow \nu + 1$

$\triangleright$ Go to next iteration.

end while

3.3 Numerical Results

The algorithm 1 has been implemented in python and the parameters have been tuned to $\gamma = 10$, $\text{max_iter} = 10\,000$, and $\epsilon = 0.0000001$. The model converges in 7391 iterations. The objective value of the solution is 4754.47 \$ which differs from the true optimal solution by 0.04 \$. The same four generators are activated as in the optimal solution and they produce the same within a margin of 0.04 MW. Thus, it is concluded that the decomposition and application of Consensus ADMM is successful.

By using a lower value of γ a solution is found using more iterations as the values change from iteration to iteration is smaller and it takes the model a longer

time to reach the right value. A greater value of γ on the other hand makes the steps bigger but the model here has a tendency to overshoot and thereby fluctuate more before finding a solution with a small enough error.

4 Multi-time Step OPF with Battery Energy Storage Systems (BESSs)

We now consider that three battery energy storage systems (BESSs) $i \in \mathcal{B} = \{1, 2, 3\}$ are placed in the power system. We define the parameters $u_{in} \in \{0, 1\}$, which is equal to 1 if battery i is placed at node n and 0 otherwise. We arbitrarily place the batteries at nodes 3, 16 and 23, which contain wind producers. Therefore, we set $u_{1,3} = u_{2,16} = u_{3,23} = 1$ and $u_{in} = 0$ for all other values. We define the initial state-of-charge $\text{SOC}_i^{\text{init}}$ of all batteries to be equal to 0.

All sets and input parameters of the multi-time step DC optimal power flow (OPF) are defined in Section 1.

4.1 Mathematical Formulation

In the multi-time step OPF with Battery Energy Storage Systems, solved over 24 hours, the primal decisions variables of the single-time step OPF problem (1a)-(1f) are extended for all hours $t \in \mathcal{T}$. Additionally, for each time step t and battery $i \in \mathcal{B}$, we introduce the following decision variables: its state-of-charge SOC_{it} (MWh), and discharging p_{int}^{dis} and charging p_{int}^{ch} power (MW) at node $n \in \mathcal{N}$.

This multi-time step OPF problem with BESS can be formulated as:

$$\min_{\Omega^P} \sum_{t \in \mathcal{T}} \left[\sum_{g \in \mathcal{G}} c_g \mathbf{p}_{gt} + \sum_{d \in \mathcal{D}} \text{VoLL} \mathbf{sh}_{dt} \right] \quad (11a)$$

$$\begin{aligned} \text{s.t. } & \sum_{g \in \mathcal{G}_n} \mathbf{p}_{gt} + \sum_{i \in \mathcal{B}} (\mathbf{p}_{int}^{\text{dis}} - \mathbf{p}_{int}^{\text{ch}}) \\ & = \sum_{d \in \mathcal{D}_n} (\bar{P}_{dt} - \mathbf{sh}_{dt}) + \sum_{m \in \mathcal{N}_n} b_{nm} (\boldsymbol{\theta}_{nt_0} - \boldsymbol{\theta}_{mt}), \forall n \in \mathcal{N}, t \in \mathcal{T} : \boldsymbol{\lambda}_{nt} \end{aligned} \quad (11b)$$

$$0 \leq \mathbf{p}_{gt} \leq \bar{P}_g^G, \forall g \in \mathcal{G}, t \in \mathcal{T} \quad (11c)$$

$$0 \leq \mathbf{sh}_{dt} \leq \bar{P}_{dt_0}^D, \forall d \in \mathcal{D}, t \in \mathcal{T} \quad (11d)$$

$$b_{nm} (\boldsymbol{\theta}_{nt} - \boldsymbol{\theta}_{mt}) \leq \bar{P}_{nm}^F, \forall n \in \mathcal{N}, m \in \mathcal{N}_n, t \in \mathcal{T} \quad (11e)$$

$$\boldsymbol{\theta}_{1t} = 0, \forall t \in \mathcal{T} \quad (11f)$$

$$\text{SOC}_{i1} = \text{SOC}_i^{(\text{init})} + \rho_i^{\text{ch}} \mathbf{p}_{i1}^{\text{ch}} - \rho_i^{\text{dis}} \mathbf{p}_{i1}^{\text{dis}}, \forall i \in \mathcal{B} \quad (11g)$$

$$\text{SOC}_{it} = \text{SOC}_{i(t-1)} + \sum_{n \in \mathcal{N}} (\rho_i^{\text{ch}} \mathbf{p}_{int}^{\text{ch}} - \rho_i^{\text{dis}} \mathbf{p}_{int}^{\text{dis}}), \forall i \in \mathcal{B}, t \in \mathcal{T} \quad (11h)$$

$$0 \leq \mathbf{p}_{int}^{\text{ch}} \leq \bar{P}_i^{\text{ch}} u_{in}, \forall i \in \mathcal{G}, n \in \mathcal{N}, t \in \mathcal{T} \quad (11i)$$

$$0 \leq \mathbf{p}_{int}^{\text{dis}} \leq \bar{P}_i^{\text{dis}} u_{in}, \forall i \in \mathcal{G}, n \in \mathcal{N}, t \in \mathcal{T} \quad (11j)$$

$$0 \leq \text{SOC}_{it} \leq \overline{\text{SOC}}_i, \forall i \in \mathcal{B}, t \in \mathcal{T}, \quad (11k)$$

$$\text{SOC}_{i24} \geq \text{SOC}_i^{(\text{init})}, \forall i \in \mathcal{B} \quad (11l)$$

where the set of decision variables of the multi-time step OPF for all hours is

$$\begin{aligned} \Omega^P = & \{\mathbf{p}_{gt}\}_{g \in \mathcal{G}, t \in \mathcal{T}} \cup \{\boldsymbol{\theta}_{nt}\}_{n \in \mathcal{N}, t \in \mathcal{T}} \cup \{\mathbf{sh}_{dt}\}_{d \in \mathcal{D}, t \in \mathcal{T}} \\ & \cup \{\text{SOC}_{it}\}_{i \in \mathcal{B}, t \in \mathcal{T}} \cup \{\mathbf{p}_{int}^{\text{ch}}, \mathbf{p}_{int}^{\text{dis}}\}_{i \in \mathcal{B}, n \in \mathcal{N}, t \in \mathcal{T}}. \end{aligned}$$

This OPF problem aims at minimizing the system operating cost (11a), consisting of electricity production cost and load shedding cost, subject to the OPF constraints, which are similar to the single-time step OPF, and expressed at each time step $t \in \mathcal{T}$, namely: power balance at each node (11b), upper- and lower-bounds on the power production of each generator (11c), load shedding of each consumer

(11d), power flow between all connected nodes (11e), and the reference voltage angle at the slack bus (11g). Note that in the balance equation (11b) batteries located at a specific node can act as producers, when discharging, or as loads, when charging. In addition, operating constraints of the batteries are considered, and include: the update of the state-of-charge at the initial time (11g) and each following time step $t \in \mathcal{T} \setminus \{1\}$ (11h), the upper and lower bound for the batteries charging and discharging power, and state of charge (11i)-(11k), and boundary conditions of the state-of-charge and the final state-of-charge (11l). Note that constraints (11i) and (11j) enforce that the charging and discharging power of each battery at a specific node is equal to 0 if this battery is not located at this node, i.e. if $u_{in} = 0$. As a result, in the SOC update constraints (11g) and (11h) for each battery, only one of the charging and discharging variables takes a non-zero value, for the node n at which the battery is connected. Similarly, in the balance equations (11b) at each node, only the charging and discharging power of the batteries located at this node are counted.

Note that the dual variable λ_{nt} associated with the balance equation represents the nodal price at node n and time t . This minimization problem is convex, because its objective function, equality and inequality constraints are all **linear**, and therefore convex.

4.2 Numerical Results

The mathematical model has been implemented in the python function `runMultiStepOPF`. The main results are presented in tabs. 3 and 4. We recall that the three batteries have been placed arbitrarily in node 3, node 16, and node 23.

The objective value of the multi-step OPF without a BESS is 160 266.74 \$ while it drops to 155 607.54 \$ when a BESS is implemented. This is a difference of 4 659.2 \$ and a reduction of the cost by approximately 3%. Even though the total cost for all generators combined is reduced, some generators face an increased production cost when implementing the BESS. This is generator G1, G2, G8, and G9. The reduction in the total cost is found at generators G11 and G12. The reduction in cost for these two generators is responsible for the total cost when looking at all generators combined.

Looking at the LMPs averages it can be seen that the average LMP for each

generator increases when implementing a BESS. This applies to all generators but G9 and G10 which keep the same LMP with or without a BESS. The reason for this increase may be due to the fact that when there is no BESS, only the exact amount of power is drawn from the generators. When the BESS is implemented the system tries to maximize the utilization of the cheapest generators and as a result, there is an increase in total production due to the battery efficiencies. However, the objective value/total system cost is improved, when the BESS is implemented. This means that the average cost pr. MWh in the system is decreased with the introduction of BESS. To further look at the LMP values, two network plots have been made with and without BESS - both are for Hour T5.

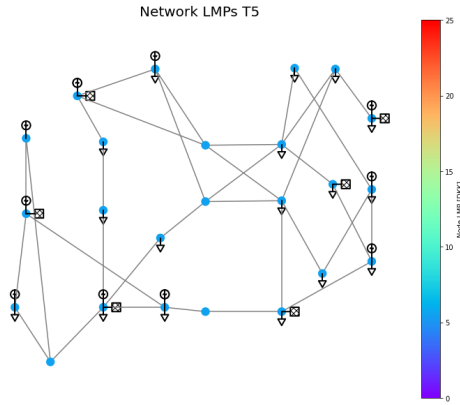


Figure 2: LMP plot without BESS

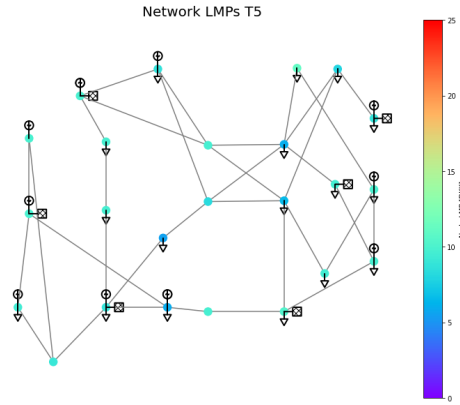


Figure 3: LMP plot with BESS

By looking at the figures above, it can be seen by the color of the nodes, that the nodes in the plot with BESS installed (fig. 3), generally have a turquoise color, which indicates a higher LMP value, than the blue color the nodes have in fig. 2. The same two plots have been made for the next hour T6. By looking at figs. 6 and 7, it is seen that most nodes have roughly the same color, but especially two nodes have a noticeably higher LMP value when the BESS is not installed. This observation indicates that while introducing BESS results in a higher average LMP, it reduces the peak LMP values. This means that for the BESS system, the LMP values fluctuate less, introducing a more stable price to the system.

Similar to the single-step OPF there is congestion in the system occurring between node 14 and 16 and between node 15 and 21. As shown in appendix D and appendix E, there is both congestion with and without the BESS. Without the BESS the congestion between node 14 and 16 occurs for 17 hours, with the BESS the congestion between these nodes occurs for 19 hours. The same applies to nodes 15 and 21 where congestion happens for 10 without BESS and for 21 hours with BESS. In total, congestion is happening for 27 hours without the BESS and 40 hours with the BESS. The reason for the increased congestion when the BESS is implemented can be found in the increased production. The production is increased in order to charge the battery and to cover the losses due to charging and discharging efficiencies. Therefore more energy can be congested in the system. Even though the congestion occurs for more hours when the BESS is implemented the dual variables associated with the maximum flow constraints are lower compared to when the BESS is not implemented. Consequently, the congestion has less influence on the objective value when the BESS is implemented and the gain which can be achieved by increasing the flow constraint limits is also lowered.

Regarding wind utilization, the production of electricity from the wind farms remains the same in the two scenarios with or without a BESS. See appendix A and B. This is due to the fact that the wind farms have a production cost of zero, thus it is always best to utilize the wind farms to their maximum capacity. This is what is done in both scenarios, where wind farms always produce at their maximum. The following two tables summarise the production, cost, profit and LMP for each generator for the scenario with and without batteries.

Table 3: Variables and parameters related to the Generators

Multi-time Step Optimal Power Flow (Generators) - without BESS												
	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
Prod sum	1777.5	2236.8	0	0	0	0	0	1425.4	8253.4	7200	4043.8	966,7
Cost	23677	29794	0	0	0	0	0	8580	45146	0	42541	10527
Profit	1666.3	1823.5	0	0	0	0	0	0	2880.5	42342.3	664.8	0
LMP avg	11.5	11.6	12	11.7	6.9	6.9	6.4	5.9	5.8	5.9	9.2	9.2

Table 4: Variables and parameters related to the Generators

Multi-time Step Optimal Power Flow (Generators) - with BESS												
	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
Prod sum	1900.3	2331.4	0	0	0	0	0	2366.9	9068.6	7200	2927.8	421.2
Cost	25312	31054	0	0	0	0	0	14248	49605	0	30799	4587
Profit	1112.3	1260.8	0	0	0	0	0	0	2771.1	42792.7	229.4	0
LMP avg	11.9	12	12.4	12.2	7.5	7.5	6.7	6	5.8	5.9	9.6	9.6

5 Optimal Placement of BESSs

As a central planner/system operator, we now consider how to optimally place these 3 BESSs in the power system in order to minimize the system's production cost. Each BESS cannot be split between multiple locations in the power system, however, multiple BESSs can be placed in the same location. As a result the storage capacity installed at each node of the system is either 0 MWh (0 BESS), 450 MWh (1 BESS), 900 MWh (2 BESSs) or 1350 MWh (3 BESSs). The system operation can be modelled over a single (representative) day.

We consider the multi-time step OPF problem defined in Section 4 and the sets, input parameters and decision variables defined in Section 1 and Section 4. In addition, we now consider that the placement of the batteries $u_{in} \in \{0, 1\}$ is a binary decision variable, which is equal to 1 if battery i is placed at node n and 0 otherwise.

5.1 Mathematical Formulation

This multi-time step OPF problem with BESS can be formulated as:

$$\min_{\Omega^{\text{BESS}}} \sum_{t \in \mathcal{T}} \left[\sum_{g \in \mathcal{G}} c_g p_{gt} + \sum_{d \in \mathcal{D}} \text{VoLL} s_{dt} \right] \quad (12a)$$

$$\text{s.t. (11b) -- (11l)} \quad (12b)$$

$$\sum_{n \in \mathcal{N}} u_{in} = 1, \forall i \in \mathcal{B}, \quad (12c)$$

where the set of decision variables $\Omega^{\text{BESS}} = \Omega^P \cup \{\mathbf{u}_{in}\}_{i \in \mathcal{B}, n \in \mathcal{N}}$ includes all the decision variables Ω^P of the multi-time step OPF problem in Section 4 and the binary variables enforcing the placement of the batteries. This optimal placement problem aims at minimizing the system operating cost over 1 typical day (12a). It is subject to the same operating constraints as the multi-time step OPF in which the placement of the batteries is a decision variable (12b). In addition, operating constraints ensuring that each battery is placed at a single node are enforced (12c). This minimization problem is non-convex, due to the binary variables $\mathbf{u}_{in}$.

5.2 Numerical Results

The mathematical model above has been implemented in the python function `runOptPlaceBESS` and results are presented in tab. 5.

By solving the optimal placement problem the objective system costs becomes 153 560.83 \$, which, compared to the arbitrary placement of the batteries, provides a reduction of 1.3%, and, compared to the system without BESSs, a reduction of approximately 4%. The optimization model chooses to place the three batteries at nodes 3, 5, and 7. All three nodes also have a load connected to them. Due to increased production in some nodes, the production cost is increased for generators G9, G11, and G12 compared to the arbitrary placement of the batteries. The production and thus the production cost is reduced for generators G1, G2, and G8. The production and the production cost remain the same for generator G10 as the maximum capacity has been reached in all scenarios. Over the course of the whole day there is a reduction in the combined production of all generators when the batteries are placed on the optimal nodes, compared to the placements in task 3.

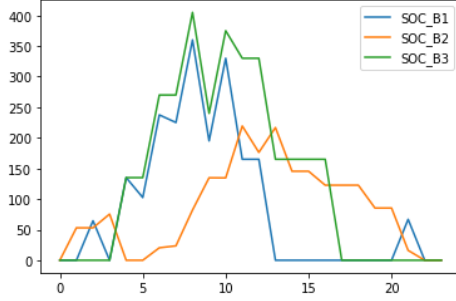


Figure 4: TASK 3 - Battery SOC

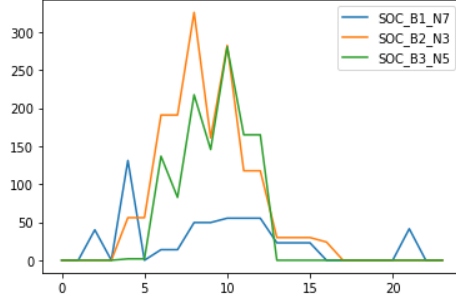


Figure 5: TASK 4 - Battery SOC

Two plots of the batteries SOC's are presented above, the placement of batteries for task 3 is described in subsection 4.2, where it can be seen that more energy is stored in the BESS in task 3, than in task 4. This decrease in stored energy means that less energy is lost due to efficiencies, but also that less energy has to be produced by the generators, which improves the objective value of the problem.

The LMPs average increased once again, this applies to all generators except G8 and G9. Here the average LMP decreases when the batteries are placed in a more optimal location. Congestion is occurring between Node 14 and 16 and between Node 15 and 21. As shown in appendix F There are 23 hours of congestion between nodes 14 and 16 and 20 hours of congestion between nodes 15 and 21. For the line between Node 14 and 16 the dual variables associated with the constraints are on average lower compared to the dual variables when the batteries are arbitrarily placed. For the line between Node 15 and 21 the dual variables are on average not lower than the dual variables when the batteries are arbitrarily placed but they are still lower than when the BESS was not implemented.

Table 5: Variables and parameters related to the Generators

Multi-time Step Optimal Power Flow BESS placement (Generators)												
	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
Prod sum	1336.1	2236.8	0	0	0	0	0	2087.6	9179.2	7200	3665.9	424.9
Cost	17797	29794	0	0	0	0	0	12567	50210	0	38565	4627
Profit	288.8	413.2	0	0	0	0	0	0	1613.5	42312.9	229.4	0
LMP avg	12.5	12.5	13	12.7	7.7	7.7	6.9	5.9	5.6	5.9	9.9	9.9

The table summarises relevant information about each generator in the scenario. It shows which generators are activated and data related to the prices in each node.

References

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- [2] W. Bukhsh, “Data for stochastic multiperiod optimal power flow problem,” 2025. [Online]. Available: <https://sites.google.com/site/datasmopf/>

A Appendix A

Multi-time Step Optimal Power Flow (Wind) - without BESS						
	W1	W2	W3	W4	W5	W6
Production sum	5033,3	4974,5	4671,8	4535,2	2899	4753,9
Total profit	49962,6	58461,7	55850,3	28707,9	16740,6	43597,5
LMP average	9,9	11,8	12	6,4	5,8	9,2

B Appendix B

Multi-time Step Optimal Power Flow (Wind) - with BESS						
	W1	W2	W3	W4	W5	W6
Production sum	5033,3	4974,5	4671,8	4535,2	2899	4753,9
Total profit	52222	60583,2	57910,5	30509,4	16709,4	45164,4
LMP average	10,4	12,3	12,4	6,7	5,8	9,6

C Appendix C

Figure 6: LMP plot without BESS hour T6

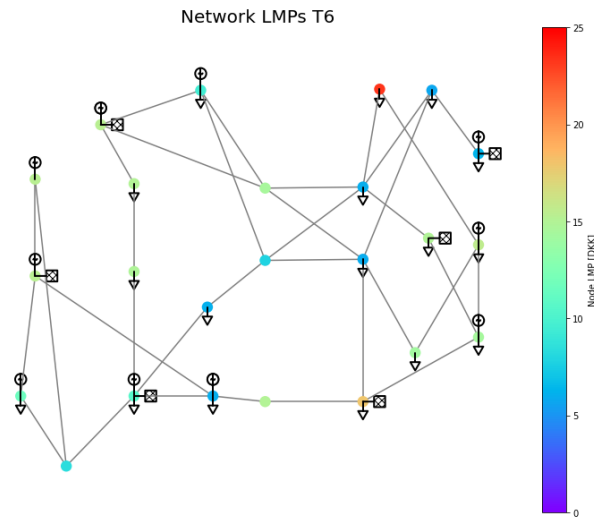
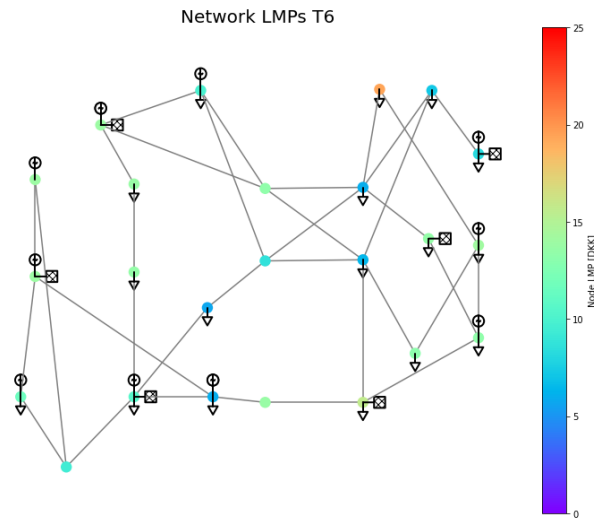


Figure 7: LMP plot with BESS Hour T6



D Appendix C

Congestion in Multi-time Step without BESS		
Time	N14 , N16	N15 , N21
T1	-13,4091	-4,75102
T2	-27,0946	-
T3	-	-
T4	-27,0946	-
T5	-	-
T6	-25,0361	-
T7	-	-
T8	-25,0361	-
T9	-	-
T10	-27,0946	-
T11	-	-
T12	-27,0946	-
T13	-16,8744	-3,33501
T14	-27,0946	-
T15	-17,2702	-3,17331
T16	-17,2702	-3,17331
T17	-17,2702	-3,17331
T18	-14,9475	-4,79167
T19	-17,2702	-3,17331
T20	-16,8744	-3,33501
T21	-17,2702	-3,17331
T22	-17,2702	-
T23	-	-3,17331
T24	-	-
Avg	-20,66	-3,53

E Appendix D

Congestion in Multi-time Step with BESS		
Time	N14 , N16	N15 , N21
T1	-13,4091	-4,75102
T2	-23,1184	-1,62476
T3	-6,42525	-4,56648
T4	-23,1184	-1,62476
T5	-	-4,39669
T6	-24,344	-0,28282
T7	-	-
T8	-24,344	-0,28282
T9	-	-
T10	-24,344	-0,28282
T11	-	-2,71696
T12	-24,344	-0,28282
T13	-17,1095	-3,16907
T14	-24,344	-0,28282
T15	-17,2702	-3,17331
T16	-17,2702	-3,17331
T17	-17,2702	-3,17331
T18	-17,2702	-3,17331
T19	-17,2702	-3,17331
T20	-17,1095	-3,16907
T21	-17,2702	-3,17331
T22	-8,41737	-2,93939
T23	-17,2702	-3,17331
T24	-	-
Avg	-18,49	-2,50

F Appendix E

Congestion in Multi-time Step Optimal BESS Placement		
Time	N14 , N16	N15 , N21
T1	-13,4091	-4,75102
T2	-27,0946	-
T3	-8,70819	-4,6268
T4	-15,3426	-4,80211
T5	-13,7056	-1,4693
T6	-17,2702	-3,17331
T7	-16,0814	-
T8	-17,2702	-3,17331
T9	-16,0814	-
T10	-17,2702	-3,17331
T11	-13,4499	-0,7903
T12	-17,2702	-3,17331
T13	-16,8744	-3,33501
T14	-17,2702	-3,17331
T15	-17,2702	-3,17331
T16	-17,2702	-3,17331
T17	-17,2702	-3,17331
T18	-17,2702	-3,17331
T19	-17,2702	-3,17331
T20	-16,8744	-3,33501
T21	-17,2702	-3,17331
T22	-8,60445	-4,62406
T23	-17,2702	-3,17331
T24	-	-
Avg	-16,24	-3,29